**University of British Columbia**
Department of Computer Science

# CPSC 304 Project Cover Page

Milestone #: 2

Date: Oct 15, 2024

Group Number: 43

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By typing our names and student numbers in the above table, we certify that the work in the attached assignment was performed solely by those whose names and student IDs are included above. (In the case of Project Milestone 0, the main purpose of this page is for you to let us know your e-mail address, and then let us assign you to a TA for your project supervisor.)
In addition, we indicate that we are fully aware of the rules and consequences of plagiarism, as set forth by the Department of Computer Science and the University of British Columbia

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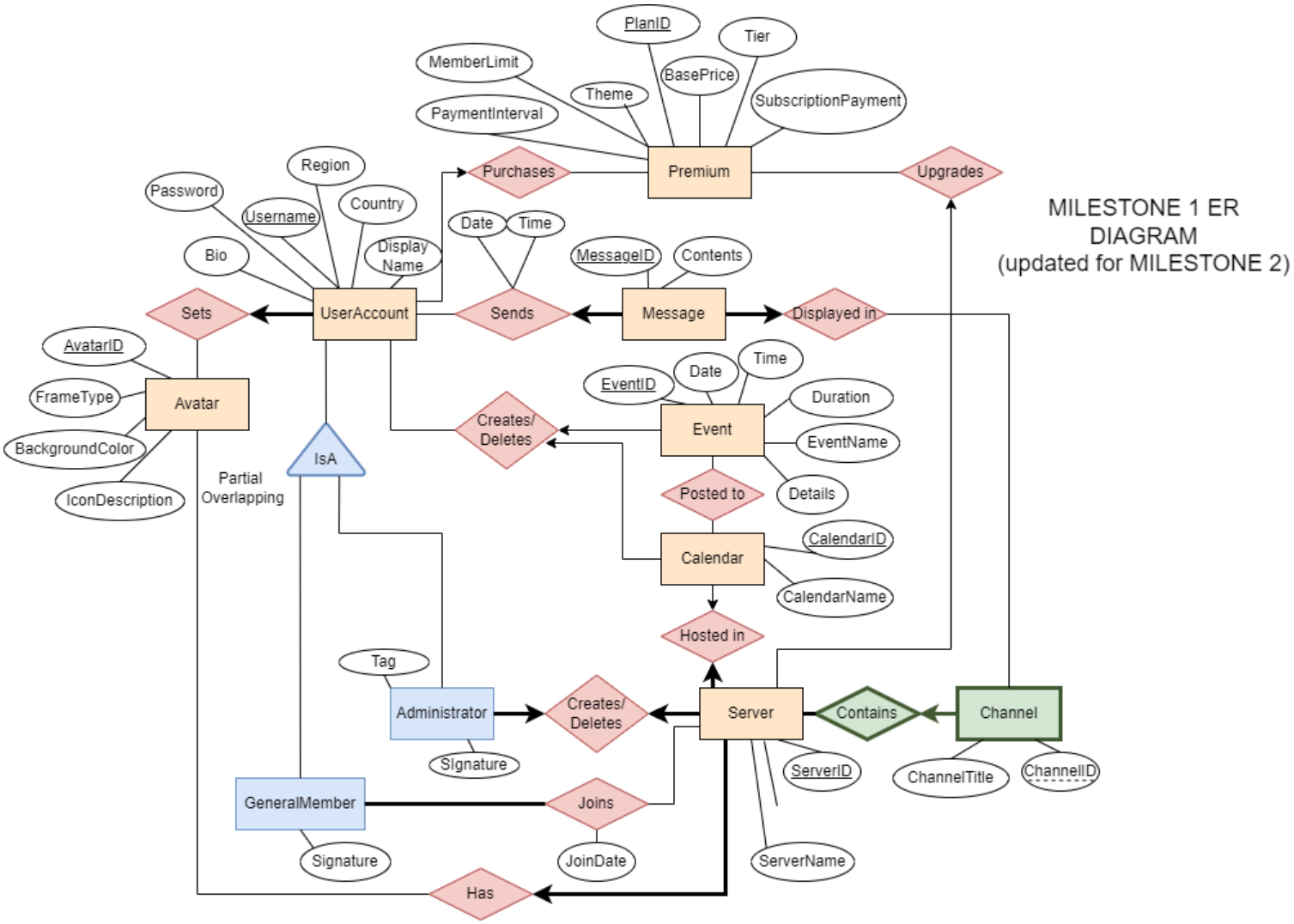
# Project Summary

Our project is a communication application modeling user interactions through messages in community spaces called servers. The application also includes premium plans, calendars, events, profile avatars, and assigning roles to accounts such as server administrator and general members.

# ER Diagram



**Changes and adjustments made to ER Diagram since Milestone #1:**

1. Addition of Country and Region attributes to UserAccount to add functional dependency.
2. Added attributes BasePrice and SubscriptionPayment to Premium entity to set up significant functional dependencies for Milestone 2 normalization steps.
3. Changed attribute from privileges into tag and signature in Administrator entity to store single-valued data types
4. Changed participation constraint between Server and Calendar to ensure Server has a one-to-one relationship with Calendar.
5. Changed participation constraint between Server and Premium from a many-to-many relationship into many-to-one.
6. Changed participation constraint between generalMember and Server into a mandatory participation.
7. Renamed relationship between administrator and server into creates/deletes, removed settings attribute from the same relationship.
8. Removed attribute DatePurchased from relationship between UserAccount and Premium for clarity. PaymentInterval attribute moved from relationship to Premium entity.
9. Removed features attribute from relationship between Server and Premium for clarity.
10. Added or renamed the following additional attributes to entities:
    a. Added Theme and MemberLimit attribute to Premium entity
    b. Renamed Icon attribute to IconDescription in Avatar entity
    c. Renamed LogID to MessageID in Message entity
11. Removed participation constraint between event and calendar so that events can exist on their own without being posted to calendars (such as default events like Stat holidays).

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# Relational Model Formal Definitions

Primary Keys: Underlined
Foreign Keys: **Bolded**

UserAccount (Username: VARCHAR[30], DisplayName: VARCHAR[30], Password: VARCHAR[50], Bio: VARCHAR[250], Country: VARCHAR[50], Region: VARCHAR[50], **AvatarID**: INTEGER [5], **PlanID**: INTEGER[10])

- Primary Key: Username
- Candidate Key: Username
- Foreign Key(s):
    - AvatarID (references Avatar)
    - PlanID (references Premium)
- Constraints: AvatarID cannot be null

Avatar (AvatarID: INTEGER [5], FrameType: VARCHAR[30], BackgroundColor: CHAR[7], IconDescription: VARCHAR[250])

- Primary Key: AvatarID
- Candidate Key: AvatarID
- Foreign Key(s): None
- Constraints: None

GeneralMember (**Username**: VARCHAR[30], Signature: VARCHAR[250])

- Primary Key: Username
- Candidate Key: Username
- Foreign Key(s): Username (references UserAccount)
- Constraints: None

Administrator (**Username**: VARCHAR[30], Tag: VARCHAR[30], Signature: VARCHAR[250], **ServerID**: INTEGER[5])

- Primary Key: Username
- Candidate Key: Username, ServerID
- Foreign Key(s): Username (references UserAccount), ServerID (references Server)
- Constraints: ServerID cannot be null and must be unique

Server (ServerID: INTEGER[5], ServerName: VARCHAR[30], **PlanID**: INTEGER[10], **CalendarID**: INTEGER[5], **AvatarID**: INTEGER [5])

- Primary Key: ServerID
- Candidate Key: ServerID

- Foreign Key(s): PlanID (references Premium), CalendarID (references Calendar), AvatarID (references Avatar),
- Constraints: CalendarID and AvatarID cannot be null, and CalendarID must be unique

Channel (<u>ChannelID</u>: INTEGER[5], ChannelTitle: VARCHAR[30], **<u>ServerID</u>**: INTEGER[5])

- Primary Key: {ChannelD, ServerID}
- Candidate Key: {ChannelD, ServerID}
- Foreign Key(s): ServerID (references Server)
- Constraints: ServerID cannot be null

Message (<u>MessageID</u>: INTEGER[10], Contents: VARCHAR[250], **Username**: CHAR[30], **ChannelID**: INTEGER[5], **ServerID**: INTEGER[5], Date: DATE, Time: TIME)

- Primary Key: MessageID
- Candidate Key: MessageID
- Foreign Key(s): Username (references UserAccount), {ChannelID, ServerID} (references Channel)
- Constraints: Username, ServerID, and ChannelID cannot be null

Premium (<u>PlanID</u>: INTEGER[10], BasePrice: INTEGER[5], Tier: VARCHAR[10], PaymentInterval: VARCHAR[10], SubscriptionPayment: INTEGER[5], MemberLimit: INTEGER[3], Theme: CHAR[7])

- Primary Key: PlanID
- Candidate Key: {PlanID}, {Tier, PaymentInterval}
- Foreign Key(s): None
- Constraints: None

Calendar (<u>CalendarID</u>: INTEGER[5], CalendarName: VARCHAR[30], **Username**: VARCHAR[30]])

- Primary Key: CalendarID
- Candidate Key: CalendarID
- Foreign Key(s): Username (references UserAccount)
- Constraints: None

Event (<u>EventID</u>: INTEGER[5], EventName: VARCHAR[30], Time: TIME, Date: DATE, Duration: INTEGER[2], Details: VARCHAR[250], **Username**: VARCHAR[30])

- Primary Key: EventID
- Candidate Key: EventID
- Foreign Key(s): Username (references UserAccount)

- Constraints: None

PostedTo (**EventID**: INTEGER[5], **CalendarID**: INTEGER[5])

- Primary Key: {EventID, CalendarID}
- Candidate Key: {EventID, CalendarID}
- Foreign Key(s): EventID (references Event), CalendarID (references Calendar)
- Constraints: None

Joins (**MemberUsername**: VARCHAR[30], **ServerID**: INTEGER[5], JoinDate: DATE)

- Primary Key: {MemberUsername, ServerID)
- Candidate Key: {MemberUsername, ServerID)
- Foreign Key(s): MemberUsername (references GeneralMember(Username)), ServerID (references Server)
- Constraints: ServerID cannot be null

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# Functional Dependencies

UserAccount (<u>Username</u>, DisplayName, Password, Bio, Country, Region, **AvatarID**, **PlanID**)
FDs:

- Username → Username, DisplayName, Password, Bio, Country, Region, AvatarID, PlanID
- Region → Country

Avatar (<u>AvatarID</u>, FrameType, BackgroundColor, IconDescription)
FDs:

- AvatarID → AvatarID, FrameType, BackgroundColor, IconDescription

GeneralMember (**<u>Username</u>**, Signature)
FDs:

- Username → Username, Signature

Administrator (**<u>Username</u>**, Tag, Signature, **ServerID**)
FDs:

- Username → Username, Tag, Signature, ServerID
- ServerID → Username

Server (<u>ServerID</u>, ServerName, **PlanID**, **CalendarID**, **AvatarID**)
FDs:

- ServerID → ServerID, ServerName, PlanID, CalendarID, AvatarID
- CalendarID → ServerID

Channel (<u>ChannelID</u>, ChannelTitle, **<u>ServerID</u>**)
FDs:

- ChannelID, ServerID → ChannelID, ChannelTitle, ServerID

Message (<u>MessageID</u>, Contents, **Username**, **ChannelID**, **ServerID**, Date, Time)
FDs:

- MessageID → MessageID, Contents, Username, ChannelID, ServerID, Date, Time

Premium (<u>PlanID</u>, BasePrice, Tier, MemberLimit, Theme, PaymentInterval, SubscriptionPayment)
FDs:

- PlanID → PlanID, BasePrice, Tier, PaymentInterval, SubscriptionPayment, MemberLimit, Theme
- PaymentInterval → SubscriptionPayment
- Tier → BasePrice
- BasePrice → Tier
- Tier, PaymentInterval → PlanID

Calendar (<u>CalendarID</u>, CalendarName, **Username**)
FDs:

- CalendarID → CalendarID, CalendarName, Username

Event (<u>EventID</u>, EventName, Time, Date, Duration, Details, **Username**)
FDs:

- EventID → EventID, EventName, Time, Date, Duration, Details, Username

PostedTo (**<u>EventID</u>**, **<u>CalendarID</u>**)

- {EventID, CalendarID} → EventID, CalendarID

Joins (**<u>MemberUsername</u>**, **<u>ServerID</u>**, JoinDate)

- {MemberUsername, ServerID} → MemberUsername, ServerID, JoinDate

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# Normalized Tables

**Normalization process for table UserAccount:**

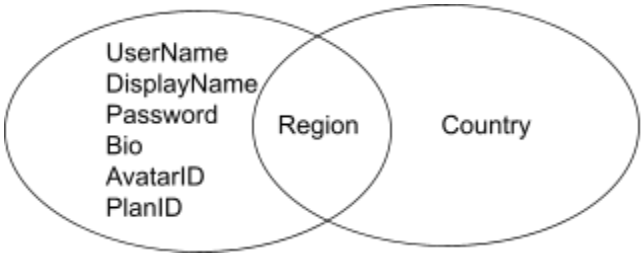
UserAccount (<u>Username</u>: VARCHAR[30], DisplayName: VARCHAR[30], Password: VARCHAR[50], Bio: VARCHAR[250], Country: VARCHAR[50], Region: VARCHAR[50], **AvatarID**: INTEGER [5], **PlanID**: INTEGER[10])
FDs:

- Username → Username, DisplayName, Password, Bio, Country, Region, AvatarID, PlanID
- Region → Country

Closures given by the FDs include:

- Username+ = {Username, DisplayName, Password, Bio, Country, Region, AvatarID, PlanID}
- Region+ = {Region, Country}

Of the closures, Username acts as a candidate key and thus a superkey for the table. The FD for Region → Country violates BCNF and 3NF because Region is not a superkey for this table. We will begin decomposing using Region → Country




**Final Tables for Normalization of UserAccount to BCNF**

1. UserAccount (<u>Username</u>: VARCHAR[30], DisplayName: VARCHAR[30], Password: VARCHAR[50], Bio: VARCHAR[250], **AvatarID**: INTEGER[5], **PlanID**: INTEGER[10], **Region**: VARCHAR[50])
    - Primary Key: Username
    - Candidate Key: Username
    - Foreign Key(s):
        i. AvatarID (references Avatar)
        ii. PlanID (references Premium)
        iii. Region (references Location)
    - Constraints: AvatarID cannot be null
2. Location (<u>Region</u>: VARCHAR[50], Country: VARCHAR[50])
    - Primary Key: Region

- Candidate Key: Region
- Foreign Key(s): None
- Constraints: None

**Normalization process for table PremiumPlan:**

Premium (<u>PlanID</u>: INTEGER[10], BasePrice: INTEGER[5], Tier: VARCHAR[10], PaymentInterval: VARCHAR[10], SubscriptionPayment: INTEGER[5], MemberLimit: INTEGER[3], Theme: CHAR[7])
FDs:

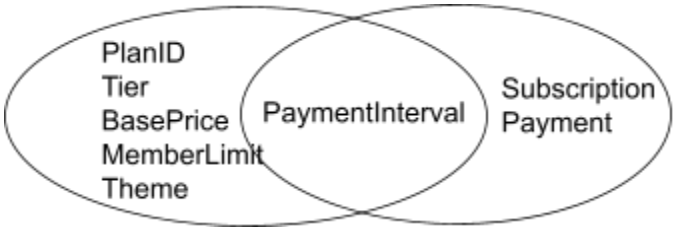
- PlanID → PlanID, BasePrice, Tier, PaymentInterval, SubscriptionPayment, MemberLimit, Theme
- PaymentInterval → SubscriptionPayment
- Tier → BasePrice
- BasePrice → Tier

From the primary key PlanID, we note that we have minimal cover FD as the following: PlanID → Tier, PaymentInterval

Closures given by the FDs include:

- PlanID+ = {PlanID, BasePrice, Tier, PaymentInterval, SubscriptionPayment, MemberLimit, Theme}
- PaymentInterval+ = {PaymentInterval, SubscriptionPayment}
- Tier+ = {Tier, BasePrice}
- BasePrice+ = {BasePrice, Tier}

Of the closures, only PlanID acts as a candidate key and thus a superkey for the table. The other FDs violate BCNF and 3NF. We will begin decomposing using PaymentInterval → SubscriptionPayment.




Resulting tables:

1. PremiumPlan(<u>PlanID</u>: INTEGER[10], Tier: VARCHAR[10], BasePrice: INTEGER[5], MemberLimit: INTEGER[3], Theme: CHAR[7], **PaymentInterval**: VARCHAR[10])

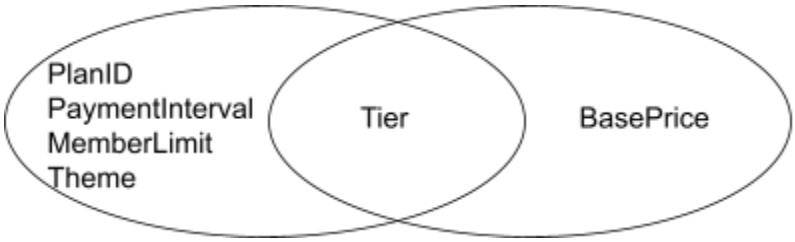
- Primary Key: PlanID
- Candidate Key: PlanID
- Foreign Key(s): PaymentInterval (references Payment)
- Constraints: None

2. Payment(<u>PaymentInterval</u>: VARCHAR[10], SubscriptionPayment: INTEGER[5])
    - Primary Key: PaymentInterval
    - Candidate Key: PaymentInterval
    - Foreign Key(s): None
    - Constraints: None

Resulting closures:

- PlanID+ = {PlanID, BasePrice, Tier, PaymentInterval, SubscriptionPayment, MemberLimit, Theme}
- PaymentInterval+ = {PaymentInterval, SubscriptionPayment}
- Tier+ = {Tier, BasePrice}
- BasePrice+ = {BasePrice, Tier}

Of the closures, the last FDs that violates BCNF and 3NF is Tier+ = {Tier, BasePrice} and BasePrice+ = {BasePrice, Tier}. We will choose one of these to decompose the table PremiumPlan on, Tier → BasePrice.




**Final Tables for Normalization of PremiumPlan to BCNF:**

1. PremiumPlan (<u>PlanID:</u> INTEGER[10], MemberLimit: INTEGER[3], Theme: CHAR[7], **Tier**: VARCHAR[10], **PaymentInterval**: VARCHAR[10])
    - Primary Key: PlanID
    - Candidate Key: PlanID
    - Foreign Key(s): Tier (references Tier), PaymentInterval (references Payment)
    - Constraints: None
2. Tier (<u>Tier:</u> VARCHAR[10], BasePrice: INTEGER[5])
    - Primary Key: Tier
    - Candidate Key: Tier
    - Foreign Key(s): None
    - Constraints: None

3. Payment (<u>PaymentInterval</u>: VARCHAR[10], SubscriptionPayment: INTEGER[5])
    - Primary Key: PaymentInterval
    - Candidate Key: PaymentInterval
    - Foreign Key(s): None
    - Constraints: None

## List of All Tables After Normalization:

1. UserAccount (<u>Username</u>: VARCHAR[30], DisplayName: VARCHAR[30], Password: VARCHAR[50], Bio: VARCHAR[250], **AvatarID**: INTEGER[5], **PlanID**: INTEGER[10], **Region**: VARCHAR[50])
    - Primary Key: Username
    - Candidate Key: Username
    - Foreign Key(s):
        - AvatarID (references Avatar)
        - PlanID (references Premium)
        - Region (references Location)
    - Constraints: AvatarID cannot be null
2. Location (<u>Region</u>: VARCHAR[50], Country: VARCHAR[50])
    - Primary Key: Region
    - Candidate Key: Region
    - Foreign Key(s): None
    - Constraints: None
3. Avatar (<u>AvatarID</u>: INTEGER [5], FrameType: VARCHAR[30], BackgroundColor: CHAR[7], IconDescription: VARCHAR[250])
    - Primary Key: AvatarID
    - Candidate Key: AvatarID
    - Foreign Key(s): None
    - Constraints: None
4. GeneralMember (**<u>Username</u>**: VARCHAR[30], Signature: VARCHAR[250])
    - Primary Key: Username
    - Candidate Key: Username
    - Foreign Key(s): Username (references UserAccount)
    - Constraints: None
5. Administrator (**<u>Username</u>**: VARCHAR[30], Tag: VARCHAR[30], Signature: VARCHAR[250], **ServerID**: INTEGER[5])
    - Primary Key: Username
    - Candidate Key: Username, ServerID
    - Foreign Key(s): Username (references UserAccount), ServerID (references Server)

- Constraints: ServerID cannot be null and must be unique

6. Server (<u>ServerID</u>: INTEGER[5], ServerName: VARCHAR[30], **PlanID**: INTEGER[10], **CalendarID**: INTEGER[5], **AvatarID**: INTEGER [5])
    - Primary Key: ServerID
    - Candidate Key: ServerID
    - Foreign Key(s): PlanID (references PremiumPlan), CalendarID (references Calendar), AvatarID (references Avatar),
    - Constraints: CalendarID and AvatarID cannot be null, CalendarID must be unique
7. Channel (<u>ChannelID</u>: INTEGER[5], ChannelTitle: VARCHAR[30], **<u>ServerID</u>**: INTEGER[5])
    - Primary Key: {ChannelD, ServerID}
    - Candidate Key: {ChannelD, ServerID}
    - Foreign Key(s): ServerID (references Server)
    - Constraints: ServerID cannot be null
8. Message (<u>MessageID</u>: INTEGER[10], Contents: VARCHAR[250], **Username**: CHAR[30], **ChannelID**: INTEGER[5], **ServerID**: INTEGER[5], Date: DATE, Time: TIME)
    - Primary Key: MessageID
    - Candidate Key: MessageID
    - Foreign Key(s): Username (references UserAccount), {ChannelID, ServerID} (references Channel)
    - Constraints: Username, ServerID, and ChannelID cannot be null
9. PremiumPlan (<u>PlanID</u>: INTEGER[10], **Tier**: VARCHAR[10], **PaymentInterval**: VARCHAR[10], MemberLimit: INTEGER[3], Theme: CHAR[7])
    - Primary Key: PlanID
    - Candidate Key: PlanID
    - Foreign Key(s): Tier (references Tier), PaymentInterval (references Payment)
    - Constraints: None
10. Tier (<u>Tier</u>: VARCHAR[10], BasePrice: INTEGER[5])
    - Primary Key: Tier
    - Candidate Key: Tier
    - Foreign Key(s): None
    - Constraints: None
11. Payment (<u>PaymentInterval</u>: VARCHAR[10], SubscriptionPayment: INTEGER[5])
    - Primary Key: {BasePrice, PaymentInterval}
    - Candidate Key: {BasePrice, PaymentInterval}
    - Foreign Key(s): None
    - Constraints: None

12. Calendar (<u>CalendarID</u>: INTEGER[5], CalendarName: VARCHAR[30], **Username**: VARCHAR[30])
    - Primary Key: CalendarID
    - Candidate Key: CalendarID
    - Foreign Key(s): Username (references UserAccount)
    - Constraints: None
13. Event (<u>EventID</u>: INTEGER[5], EventName: VARCHAR[30], Time: TIME, Date: DATE, Duration: INTEGER[2], Details: VARCHAR[250], **Username**: VARCHAR[30])
    - Primary Key: EventID
    - Candidate Key: EventID
    - Foreign Key(s): Username (references UserAccount)
    - Constraints: None
14. PostedTo (**<u>EventID</u>**: INTEGER[5], **<u>CalendarID</u>**: INTEGER[5])
    - Primary Key: {EventID, CalendarID}
    - Candidate Key: {EventID, CalendarID}
    - Foreign Key(s): EventID (references Event), CalendarID (references Calendar)
    - Constraints: None
15. Joins (**<u>MemberUsername</u>**: VARCHAR[30], **<u>ServerID</u>**: INTEGER[5], JoinDate: DATE)
    - Primary Key: {MemberUsername, SeverID)
    - Candidate Key:  {MemberUsername, SeverID)
    - Foreign Key(s): MemberUsername (references GeneralMember(Username)), ServerID (references Server)
    - Constraints: ServerID cannot be null

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# SQL DDL Statements

```
CREATE TABLE Payment(
        PaymentInterval         INTEGER,
        SubscriptionPayment     INTEGER,
        PRIMARY KEY (PaymentInterval)
        );

CREATE TABLE Tier(
        Tier                    VARCHAR(10),
        BasePrice               INTEGER,
        PRIMARY KEY (Tier)
        );

CREATE TABLE PremiumPlan(
        PlanID                  INTEGER,
        Tier                    VARCHAR(10),
        PaymentInterval         INTEGER,
        MemberLimit             INTEGER,
        Theme                   CHAR(7),
        PRIMARY KEY (PlanID),
        FOREIGN KEY (PaymentInterval) REFERENCES Payment,
        FOREIGN KEY (Tier) REFERENCES Tier
        );

CREATE TABLE Location(
        Country                 VARCHAR(30),
        Region                  VARCHAR(50),
        PRIMARY KEY (Region)
        );

CREATE TABLE Avatar(
        AvatarID                INTEGER,
        FrameType               VARCHAR(30),
        BackgroundColor         CHAR(7),
        IconDescription         VARCHAR(250),
        PRIMARY KEY (AvatarID)
        );
```

```
CREATE TABLE UserAccount(
        Username                VARCHAR(30),
        DisplayName             VARCHAR(30),
        Password                VARCHAR(50),
        Bio                     VARCHAR(250),
        Region                  VARCHAR(50),
        AvatarID                INTEGER NOT NULL,
        PlanID                  INTEGER,
        PRIMARY KEY (Username),
        FOREIGN KEY (Region) REFERENCES Location(Region),
        FOREIGN KEY (AvatarID) REFERENCES Avatar,
        FOREIGN KEY (PlanID) REFERENCES PremiumPlan
        );


CREATE TABLE Calendar(
        CalendarID              INTEGER,
        CalendarName            VARCHAR(30),
        Username                VARCHAR(30),
        PRIMARY KEY (CalendarID),
        FOREIGN KEY (Username) REFERENCES UserAccount
        );

CREATE TABLE Event(
        EventID                 INTEGER,
        EventName               VARCHAR(30),
        Time                    TIME,
        Date                    DATE,
        Duration                INTEGER,
        Details                 VARCHAR(250),
        Username                VARCHAR(30),
        PRIMARY KEY (EventID),
        FOREIGN KEY (UserName) REFERENCES UserAccount
        );

CREATE TABLE Server(
        ServerID                INTEGER,
        ServerName              VARCHAR(30),
        PlanID                  INTEGER,
        CalendarID              INTEGER UNIQUE NOT NULL,
```

```
        AvatarID                INTEGER NOT NULL,
        PRIMARY KEY (ServerID),
        FOREIGN KEY (PlanID) REFERENCES PremiumPlan,
        FOREIGN KEY (CalendarID) REFERENCES Calendar,
        FOREIGN KEY (AvatarID) REFERENCES Avatar,
        );

CREATE TABLE Channel(
        ChannelID               INTEGER,
        ChannelTitle            VARCHAR(30),
        ServerID                INTEGER NOT NULL,
        PRIMARY KEY (ChannelID, ServerID),
        FOREIGN KEY (ServerID)
                REFERENCES Server
                ON DELETE CASCADE
        );

CREATE TABLE GeneralMember(
        Username                VARCHAR(30),
        Signature               VARCHAR(250),
        PRIMARY KEY (Username),
        FOREIGN KEY (Username) REFERENCES UserAccount
        );

CREATE TABLE Administrator(
        Username                VARCHAR(30),
        Tag                     VARCHAR(30),
        Signature               VARCHAR(250),
        ServerID                INTEGER UNIQUE NOT NULL,
        PRIMARY KEY (Username),
        FOREIGN KEY (Username) REFERENCES UserAccount,
        FOREIGN KEY (ServerID)
                REFERENCES Server
        );

CREATE TABLE Message(
        MessageID               INTEGER,
        Contents                VARCHAR(250),
        ChannelID               INTEGER NOT NULL,
        ServerID                INTEGER NOT NULL,
```

```
    Date                    DATE,
    Time                    TIME,
    Username                VARCHAR(30) NOT NULL,
    PRIMARY KEY (MessageID),
    FOREIGN KEY (Username)
        REFERENCES UserAccount,
    FOREIGN KEY (ChannelID, ServerID) REFERENCES Channel
    );

CREATE TABLE PostedTo(
    CalendarID              INTEGER,
    EventID                 INTEGER,
    PRIMARY KEY (EventID, CalendarID),
    FOREIGN KEY (CalendarID) REFERENCES Calendar,
    FOREIGN KEY (EventID) REFERENCES Event
    );

CREATE TABLE Joins (
    MemberUsername          VARCHAR(30),
    ServerID                INTEGER NOT NULL,
    JoinDate                DATE,
    PRIMARY KEY (MemberUsername, ServerID),
    FOREIGN KEY (MemberUsername)
        REFERENCES GeneralMember (Username),
    FOREIGN KEY (ServerID) REFERENCES Server
    );
```

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# SQL DDL Inserts

--Payment

INSERT INTO Payment(PaymentInterval, SubscriptionPayment) VALUES (1, 1);

INSERT INTO Payment(PaymentInterval, SubscriptionPayment) VALUES (6, 1);

INSERT INTO Payment(PaymentInterval, SubscriptionPayment) VALUES (12, 12);

INSERT INTO Payment(PaymentInterval, SubscriptionPayment) VALUES (24, 120);

INSERT INTO Payment(PaymentInterval, SubscriptionPayment) VALUES (18, 180);

--Tier

INSERT INTO Tier(Tier, BasePrice) VALUES ("Basic", 12);

INSERT INTO Tier(Tier, BasePrice) VALUES ("Silver", 24);

INSERT INTO Tier(Tier, BasePrice) VALUES ("Gold", 60);

INSERT INTO Tier(Tier, BasePrice) VALUES ("Platinum", 120);

INSERT INTO Tier(Tier, BasePrice) VALUES ("Royal", 180);

--PremiumPlan

INSERT INTO PremiumPlan(PlanID, Tier, PaymentInterval, MemberLimit, Theme)
VALUES (1, "Basic", 1, 10, "#000000");

INSERT INTO PremiumPlan(PlanID, Tier, PaymentInterval, MemberLimit, Theme)
VALUES (2, "Silver", 1, 50, "#111111");

INSERT INTO PremiumPlan(PlanID, Tier, PaymentInterval, MemberLimit, Theme)
VALUES (3, "Gold", 6, 100, "#000000");

INSERT INTO PremiumPlan(PlanID, Tier, PaymentInterval, MemberLimit, Theme)
VALUES (4, "Platinum", 12, 150, "#FFFFFF");

```
INSERT INTO PremiumPlan(PlanID, Tier, PaymentInterval, MemberLimit, Theme)
VALUES (5, "Royal", 12, 200, "#CCCCCC");

INSERT INTO PremiumPlan(PlanID, Tier, PaymentInterval, MemberLimit, Theme)
VALUES (11, "Basic", 6, 50, "#FFFFFF");

INSERT INTO PremiumPlan(PlanID, Tier, PaymentInterval, MemberLimit, Theme)
VALUES (12, "Basic", 12, 60, "#123456");

INSERT INTO PremiumPlan(PlanID, Tier, PaymentInterval, MemberLimit, Theme)
VALUES (21, "Gold", 6, 150, "#AAAAAA");
```

--Location

```
INSERT INTO Location(Country, Region) VALUES ("Canada", "Vancouver");

INSERT INTO Location(Country, Region) VALUES ("Canada", "Ottawa");

INSERT INTO Location(Country, Region) VALUES ("USA", "Virginia");

INSERT INTO Location(Country, Region) VALUES ("India", "Odisha");

INSERT INTO Location(Country, Region) VALUES ("Canada", "Burnaby");
```

--Avatar

```
INSERT INTO Avatar(AvatarID, FrameType, BackgroundColor, IconDescription)
VALUES (1, "Circle", "#000000", "Black Circle");

INSERT INTO Avatar(AvatarID, FrameType, BackgroundColor, IconDescription)
VALUES (2, "Triangle", "#123456", "Incremental Blue Triangle");

INSERT INTO Avatar(AvatarID, FrameType, BackgroundColor, IconDescription)
VALUES (3, "Circle", "#DDDDDD", "Steam Square");

INSERT INTO Avatar(AvatarID, FrameType, BackgroundColor, IconDescription)
VALUES (4, "Rectangle", "#1247A8", "Super Silver Rectangle");
```

INSERT INTO Avatar(AvatarID, FrameType, BackgroundColor, IconDescription)
VALUES (5, "Rectangle", "#FFFFFF", "White Rectangle");

INSERT INTO Avatar(AvatarID, FrameType, BackgroundColor, IconDescription)
VALUES (10, "Circle", "#FFFFFF", "White Circle");

--UserAccount

INSERT INTO UserAccount(Username, DisplayName, Password, Bio, Region, AvatarID, PlanID)
VALUES ("ali123", "Allison Ko", "aliko123lovesdogs", "Hi, I'm Ali and I love dogs. I'm from UBC and I'm studying Computer Science, currently enrolled in a databases course with Dr. Rachel Pottinger.", "Vancouver", 1, 1);

INSERT INTO UserAccount(Username, DisplayName, Password, Bio, Region, AvatarID, PlanID)
VALUES ("ananyam", "Ananya Mahapatra", "shdie38uwnjfvjvkewfjwiw03nl", "i like lurking", "Vancouver", 3, NULL);

INSERT INTO UserAccount(Username, DisplayName, Password, Bio, Region, AvatarID, PlanID)
VALUES ("racekar", "Karen Li", "hotpotatoes", "Hello! I am very excited to be here. I like goldfish, books and hiking.", "Ottawa", 1, 5);

INSERT INTO UserAccount(Username, DisplayName, Password, Bio, Region, AvatarID, PlanID)
VALUES ("dylan_x", NULL, "hotpotatoes", NULL, "Burnaby", 3, 21);

INSERT INTO UserAccount(Username, DisplayName, Password, Bio, Region, AvatarID, PlanID)
VALUES ("goldfishlover", "Tom Thomas", "fishyfishyfishy", "I LOVE GOLDFISH", "Virginia", 5, NULL);

INSERT INTO UserAccount(Username, DisplayName, Password, Bio, Region, AvatarID, PlanID)
VALUES ("abcdefg", "...", "hijklmnop", "lalalalala", "Odisha", 1, 5);

--Calendar

INSERT INTO Calendar(CalendarID, CalendarName, Username) VALUES (1, "Dog Lovers Calendar", NULL);

INSERT INTO Calendar(CalendarID, CalendarName, Username) VALUES (2, "Recipes Calender", NULL);

INSERT INTO Calendar(CalendarID, CalendarName, Username) VALUES (3, "Socials", "ananyam");

INSERT INTO Calendar(CalendarID, CalendarName, Username) VALUES (7, "Appointments", "ananyam");

INSERT INTO Calendar(CalendarID, CalendarName, Username) VALUES (4, "Friends", "ananyam");

INSERT INTO Calendar(CalendarID, CalendarName, Username) VALUES (5, "goldfish calendar", NULL);

INSERT INTO Calendar(CalendarID, CalendarName, Username) VALUES (20, "Rock Concert Calendar", NULL);

INSERT INTO Calendar(CalendarID, CalendarName, Username) VALUES (21, "Gothic Fantasy Meetings", NULL);

INSERT INTO Calendar(CalendarID, CalendarName, Username) VALUES (10, "homework", "racekar");

--Event

INSERT INTO Event(EventID, EventName, Time, Date, Duration, Details, Username) VALUES (2, "Friendsgiving", "03:30:00", "2024-10-10", NULL, "Sara’s house", "ananyam");

INSERT INTO Event(EventID, EventName, Time, Date, Duration, Details, Username) VALUES (1, "Marley & Me Viewing", "11:00:00", "2025-01-03", 2, "Grab your blankets and tissues for a viewing of Marley & Me! It will be through Zoom and the link will be shared two days before the event. Please respond to Tara’s message in the socials channel if you will be joining.", "ali123");

INSERT INTO Event(EventID, EventName, Time, Date, Duration, Details, Username) VALUES (3, "Christmas", "00:00:00", "2024-12-25", 24, "Christmas Holiday", NULL);

INSERT INTO Event(EventID, EventName, Time, Date, Duration, Details, Username) VALUES (4, "Book Review 1", "10:30:00", "2024-12-03", 3, "Details will be posted shortly", "abcdefg");

INSERT INTO Event(EventID, EventName, Time, Date, Duration, Details, Username) VALUES (5, "Book Review 2", "10:30:00", "2024-12-27", 3, "Details will be posted shortly", "racekar");

--Server

INSERT INTO Server(ServerID, ServerName, PlanID, CalendarID, AvatarID) VALUES (1, "Dog Lovers Unite", 1, 1, 2);

INSERT INTO Server(ServerID, ServerName, PlanID, CalendarID, AvatarID) VALUES (2, "recipes", NULL, 2, 2);

INSERT INTO Server(ServerID, ServerName, PlanID, CalendarID, AvatarID) VALUES (3, "golfishies", 5, 5, 5);

INSERT INTO Server(ServerID, ServerName, PlanID, CalendarID, AvatarID) VALUES (4, "rock bands", 21, 20, 3);

INSERT INTO Server(ServerID, ServerName, PlanID, CalendarID, AvatarID) VALUES (5, "gothic fantasy", 5, 21, 4);

--Channel

INSERT INTO Channel(ChannelID, ChannelTitle, ServerID) VALUES (1, "Labradors", 1);

INSERT INTO Channel(ChannelID, ChannelTitle, ServerID) VALUES (2, "Bulldogs", 1);

INSERT INTO Channel(ChannelID, ChannelTitle, ServerID) VALUES (1, "Vegetarian", 2);

INSERT INTO Channel(ChannelID, ChannelTitle, ServerID) VALUES (1, "Breeding", 3);

INSERT INTO Channel(ChannelID, ChannelTitle, ServerID) VALUES (2, "Feeding", 3);

INSERT INTO Channel(ChannelID, ChannelTitle, ServerID) VALUES (3, "Subspecies", 3);

INSERT INTO Channel(ChannelID, ChannelTitle, ServerID) VALUES (1, "80s", 4);

INSERT INTO Channel(ChannelID, ChannelTitle, ServerID) VALUES (1, "Book of the month", 5);

INSERT INTO Channel(ChannelID, ChannelTitle, ServerID) VALUES (2, "Goodreads account sharing", 5);

--GeneralMember

INSERT INTO GeneralMember(Username, Signature) VALUES ("ali123", "all things dogs!");

INSERT INTO GeneralMember(Username, Signature) VALUES ("ananyam", "happy");

INSERT INTO GeneralMember(Username, Signature) VALUES ("racekar", "Love from Ottawa");

INSERT INTO GeneralMember(Username, Signature) VALUES ("dylan_x", NULL);

INSERT INTO GeneralMember(Username, Signature) VALUES ("goldfishlover", "there’s plenty of fish in the sea…");

INSERT INTO GeneralMember(Username, Signature) VALUES ("abcdefg", "yo!");

--Administrator

INSERT INTO Administrator(Username, Tag, Signature, ServerID) VALUES ("ali123", "admin", "all things dogs!", 1);

INSERT INTO Administrator(Username, Tag, Signature, ServerID) VALUES ("ananyam", "Big Brother", "happy", 2);

INSERT INTO Administrator(Username, Tag, Signature, ServerID) VALUES ("racekar", "admin", "be nice", 3);

INSERT INTO Administrator(Username, Tag, Signature, ServerID) VALUES ("dylan_x", "moderator", NULL, 4);

INSERT INTO Administrator(Username, Tag, Signature, ServerID) VALUES ("abcdefg", "Big Brother", "so glad you are here :)", 5);

--Message

INSERT INTO Message(MessageID, Contents, ChannelID, ServerID, Date, Time, Username) VALUES (1, "hello", 1, 1, "2020-07-01", "03:03:22", "ali123");

INSERT INTO Message(MessageID, Contents, ChannelID, ServerID, Date, Time, Username) VALUES (2, "welcome to the labrador channel", 1, 1, "2020-07-01", "03:03:55", "ali123");

INSERT INTO Message(MessageID, Contents, ChannelID, ServerID, Date, Time, Username) VALUES (33, "welcome to the bulldog channel", 2, 1, "2020-07-01", "03:05:01", "ali123");

INSERT INTO Message(MessageID, Contents, ChannelID, ServerID, Date, Time, Username) VALUES (1166, "been looking for vegetarian lasagna recipes everywhere!", 1, 2, "2024-10-10", "05:06:06", "ali123");

INSERT INTO Message(MessageID, Contents, ChannelID, ServerID, Date, Time, Username) VALUES (1169, "the ones people post here are LEGIT!", 1, 2, "2024-10-10", "05:07:22", "ananyam");

INSERT INTO Message(MessageID, Contents, ChannelID, ServerID, Date, Time, Username) VALUES (139, "are we ready to vote on book of the month?", 1, 5, "2024-08-10", "12:39:46", "racekar");

--PostedTo

INSERT INTO PostedTo(CalendarID, EventID) VALUES (1, 1);

INSERT INTO PostedTo(CalendarID, EventID) VALUES (3, 2);

INSERT INTO PostedTo(CalendarID, EventID) VALUES (4, 2);

INSERT INTO PostedTo(CalendarID, EventID) VALUES (21, 4);

INSERT INTO PostedTo(CalendarID, EventID) VALUES (21, 5);

--Joins

INSERT INTO Joins(MemberUsername, ServerID, JoinDate) VALUES ("ali123", 2, "2023-05-10");

INSERT INTO Joins(MemberUsername, ServerID, JoinDate) VALUES ("ali123", 3, "2024-01-09");

INSERT INTO Joins(MemberUsername, ServerID, JoinDate) VALUES ("ananyam", 1, "2023-09-10");

INSERT INTO Joins(MemberUsername, ServerID, JoinDate) VALUES ("racekar", 1, "2024-08-08");

INSERT INTO Joins(MemberUsername, ServerID, JoinDate) VALUES ("racekar", 2, "2024-08-08");

INSERT INTO Joins(MemberUsername, ServerID, JoinDate) VALUES ("racekar", 5, "2024-08-09");

INSERT INTO Joins(MemberUsername, ServerID, JoinDate) VALUES ("dylan_x", 2, "2023-12-12");

INSERT INTO Joins(MemberUsername, ServerID, JoinDate) VALUES ("abcdefg", 1, "2022-07-03");

INSERT INTO Joins(MemberUsername, ServerID, JoinDate) VALUES ("abcdefg", 4, "2023-01-06");

**University of British Columbia**
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